Threat Hunting and Incident Response using Wazuh SIEM

1. INTRODUCTION

Security Operations Centers (SOC) play a critical role in detecting, analyzing, and responding to cybersecurity threats. With the increasing number of cyberattacks, organizations must adopt proactive approaches such as threat hunting and automated incident response.
This report presents a comprehensive analysis of a simulated SSH brute-force attack using Wazuh SIEM. It includes threat hunting methodologies, SOAR automation, alert triage, evidence analysis, adversary emulation, and post-incident analysis.

2. OBJECTIVE

The main objectives of this task are:

- To perform threat hunting using Wazuh SIEM
- To detect suspicious activities in system logs
- To design a SOAR playbook for automated response
- To analyze alerts and classify their severity
- To perform evidence-based investigation
- To simulate attacker behavior
- To conduct post-incident analysis
- To evaluate SOC performance using metrics

3. TOOLS USED

- **Wazuh SIEM** – Log analysis and threat detection
- **Linux Terminal** – Attack simulation
- **SSH Protocol** – Used for brute-force simulation

4. THREAT HUNTING METHODOLOGY

4.1 Overview

Threat hunting is a proactive cybersecurity approach that involves actively searching for potential threats that may bypass traditional security mechanisms such as firewalls and intrusion detection systems. Unlike reactive security, where alerts trigger investigations, threat hunting focuses on identifying suspicious patterns and behaviors before they escalate into serious incidents.

In a Security Operations Center (SOC), threat hunting plays a critical role in improving detection capabilities and reducing dwell time of attackers. It involves analyzing logs, identifying anomalies, and correlating events across multiple data sources.

4.2 Hypothesis

A hypothesis-driven approach was used in this task. The following hypothesis was defined:

"Multiple failed SSH login attempts from a single source may indicate a brute-force attack."

This hypothesis is based on common attacker behavior where automated scripts attempt multiple password combinations to gain unauthorized access to a system.

4.3 Hunting Process

The threat hunting process was carried out using Wazuh SIEM. The Discover section was used to analyze system logs collected from the Linux authentication log file.

The following query was used:

rule.id:5710

This query filters events related to SSH authentication failures. The logs were carefully analyzed to identify patterns such as repeated login attempts, invalid usernames, and consistent source IP addresses.

Additional filtering techniques were applied to focus on specific fields such as:

- data.srcuser
- data.srcip
- rule.firedtimes

4.4 Findings

The analysis of logs revealed several suspicious activities:

- Multiple failed login attempts were detected within a short time interval
- Invalid usernames such as *fakeuser* and *wronguser* were repeatedly used
- All attempts originated from the same source IP address (::1)

- The value of rule.firedtimes increased from 1 to 4, indicating repeated attempts
- The attack pattern showed automated behavior consistent with brute-force techniques

These findings clearly indicate malicious activity rather than normal user behavior.

4.5 MITRE ATT&CK Mapping

The identified attack behavior was mapped to the MITRE ATT&CK framework as follows:

- **T1110.001 – Password Guessing**
  This technique involves attempting multiple passwords to gain access to an account.
- **T1021.004 – SSH**
  This technique represents remote access via SSH, which is commonly targeted by attackers.

The mapping helps in understanding attacker tactics, techniques, and procedures (TTPs), and aligns the findings with industry-standard threat intelligence.

4.6 Conclusion

Based on the analysis, it is concluded that the system experienced a brute-force attack targeting SSH services. The repeated failed login attempts, use of invalid usernames, and consistent source IP confirm malicious intent.

Threat hunting using Wazuh SIEM successfully identified the attack pattern and provided valuable insights into system vulnerabilities. This highlights the importance of proactive monitoring and log analysis in detecting early-stage attacks.

5. SOAR PLAYBOOK

5.1 Overview

Security Orchestration, Automation, and Response (SOAR) is an advanced cybersecurity solution designed to automate and streamline security operations. It integrates multiple tools such as SIEM, EDR, and ticketing systems to enable faster and more efficient incident response.

In a SOC environment, analysts often deal with a large number of alerts. SOAR reduces manual workload by automating repetitive tasks such as alert validation, IP blocking, and incident reporting. This ensures quicker response times and minimizes human error.

In this task, a SOAR playbook was designed using Wazuh SIEM alerts to simulate automated response to a brute-force attack.

5.2 Trigger Event

The playbook is triggered when a suspicious event is detected in Wazuh. In this case, the trigger condition is based on SSH authentication failure logs.

The following alert was used:

- Rule ID: 5710
- Description: SSH authentication failure
- Log Source: /var/log/auth.log

The alert indicates multiple failed login attempts using invalid usernames such as *fakeuser* and *wronguser*. Additionally, the increasing value of rule.firedtimes confirms repeated login attempts, which is a strong indicator of a brute-force attack.

5.3 Playbook Workflow

The SOAR playbook follows a structured workflow to handle the detected incident:

1. **Detect Alert**
   Wazuh SIEM detects multiple failed login attempts and generates an alert based on rule.id:5710.
2. **Extract Source IP**
   The system extracts the source IP address (::1) from the log data.
3. **Validate Repeated Attempts**
   The system checks the rule.firedtimes value to confirm multiple login failures.
4. **Classify Attack**
   Based on the pattern, the activity is classified as a brute-force attack.
5. **Block IP (Simulated)**
   The suspicious IP address is blocked to prevent further attempts (simulated action).
6. **Create Incident Ticket**
   A ticket is generated to document the incident for further investigation.
7. **Notify Administrator**
   The administrator is notified about the detected threat for immediate action.

5.4 Practical Implementation

The SOAR playbook was implemented using real Wazuh alerts generated from the simulated SSH brute-force attack. When multiple failed login attempts were detected, the system analyzed the logs and identified the source IP and repeated authentication failures.

Although a full SOAR automation tool was not used, the workflow simulates real-world automated incident response. Each step in the playbook represents how a SOC system would handle such an attack in a production environment.
The use of rule.firedtimes as a condition ensures that only repeated suspicious activities trigger the response, reducing false positives and improving accuracy.

5.5 Benefits of SOAR Implementation

The implementation of SOAR playbooks provides several advantages:

- Faster Response Time: Automated workflows reduce the time required to respond to incidents
- Reduced Manual Effort: Eliminates repetitive tasks for SOC analysts
- Improved Consistency: Ensures standardized response procedures
- Enhanced Detection Accuracy: Reduces false positives through validation steps
- Scalability: Can handle large volumes of alerts efficiently

5.6 Conclusion

The SOAR playbook successfully demonstrates how automated workflows can be used to respond to security incidents such as SSH brute-force attacks. By integrating detection and response, the system ensures faster and more efficient handling of threats.
This practical implementation highlights the importance of automation in modern SOC operations and shows how even simulated workflows can improve security posture.

6. ALERT TRIAGE

6.1 Overview

Alert triage is a fundamental process in Security Operations Centers (SOC) that involves analyzing, validating, and prioritizing security alerts generated by monitoring systems such as SIEM tools. Since modern security systems generate a large volume of alerts, it is essential to distinguish between false positives and real security incidents.
The primary goal of alert triage is to ensure that critical threats are identified quickly and addressed with appropriate response actions. Effective triage reduces alert fatigue, improves response efficiency, and enhances overall security posture.

6.2 Alert Details

The alert analyzed in this task was generated by Wazuh SIEM based on SSH authentication failure events. The following details were extracted from the logs:

- **Alert Name:** SSH Failed Login Attempt
- **Rule ID:** 5710
- **User:** fakeuser
- **Source IP:** ::1
- **Log Source:** /var/log/auth.log

The alert indicates that an invalid user attempted to log in to the system multiple times using SSH, which is not typical behavior for legitimate users.

6.3 Priority Classification

Based on the analysis, the alert is classified as:

Priority: High

This classification reflects the severity and potential risk associated with the detected activity.

6.4 Reason for Classification

The alert was classified as high priority due to the following reasons:

- **Multiple Failed Attempts:**
  The logs show repeated login failures within a short time period, indicating persistent attack behavior.
- **Invalid Usernames:**
  The use of usernames such as *fakeuser* suggests unauthorized access attempts rather than legitimate user activity.
- **Brute-Force Pattern:**
  The increasing value of rule.firedtimes confirms that multiple login attempts were made sequentially, which is a common characteristic of brute-force attacks.
- **Consistent Source IP:**
  All attempts originated from the same IP address (::1), indicating a single source performing repeated login attempts.

These factors collectively indicate malicious intent and justify a high-priority classification.

6.5 Triage Analysis

A detailed analysis of the alert reveals that the activity is not accidental but intentional and automated. The pattern of repeated login attempts with invalid credentials strongly suggests a brute-force attack. The correlation between multiple logs, consistent IP address, and repeated failures confirms that the system is being actively targeted. Such attacks, if successful, can lead to unauthorized access and compromise of system resources.

6.6 Response Recommendation

Based on the triage analysis, the following actions are recommended:

- Block the source IP address to prevent further attempts
- Implement account lockout mechanisms
- Strengthen password policies
- Monitor similar login attempts continuously

These actions help mitigate the risk and prevent escalation of the attack.

6.7 Conclusion

The alert represents a genuine and high-risk security incident. Proper triage allowed accurate identification of the threat and enabled appropriate response planning.

This process highlights the importance of alert triage in prioritizing incidents and ensuring efficient SOC operations.

7. EVIDENCE ANALYSIS

7.1 Evidence Source

The primary evidence for this investigation was collected from system authentication logs. These logs are stored in the Linux system file:

/var/log/auth.log

The logs were accessed and analyzed through Wazuh SIEM, which provides centralized log monitoring and analysis capabilities. This ensures that all relevant events are captured, processed, and correlated effectively.

7.2 Evidence Details

The collected logs contain multiple indicators of suspicious activity, including:

- Repeated failed SSH login attempts
- Use of invalid usernames such as *fakeuser* and *wronguser*
- Source IP address identified as (::1)
- Multiple login attempts within a short time frame
- Incremental increase in rule.firedtimes value (from 1 to 4)
- Authentication failure messages recorded by the SSH service

These details collectively indicate abnormal login behavior that deviates from normal system usage.

7.3 Evidence Analysis

A detailed analysis of the collected logs reveals a clear pattern of unauthorized access attempts. The repeated login failures suggest that an attacker is attempting to guess valid credentials using multiple password combinations.

The presence of invalid usernames indicates that the attacker does not have prior knowledge of valid user accounts and is attempting random guesses. This behavior is commonly associated with brute-force attacks.

Furthermore, the increasing value of rule.firedtimes confirms that the login attempts are continuous and automated rather than isolated incidents. The short time interval between attempts strengthens the conclusion that the activity is scripted.

The logs also include MITRE ATT&CK mappings such as **T1110.001 (Password Guessing)**, which further validates the classification of the attack.

7.4 Chain of Custody

To ensure the integrity and reliability of the evidence, a proper chain of custody was maintained throughout the investigation:

- **Evidence Type:** Authentication logs
- **Source:** Wazuh SIEM
- **Original Location:** /var/log/auth.log
- **Collected By:** SOC Analyst
- **Date of Collection:** April 2026
- **Integrity Status:** Maintained without modification

Maintaining a chain of custody ensures that the evidence remains credible and can be used for further analysis or reporting.

7.5 Correlation with Other Findings

The evidence collected aligns with the findings from threat hunting and alert triage. The repeated login attempts, consistent source IP, and increasing rule.firedtimes value confirm the presence of a brute-force attack.

This correlation strengthens the validity of the investigation and ensures that conclusions are based on multiple supporting data points.

7.6 Conclusion

The analysis of authentication logs clearly confirms that the system was targeted by a brute-force attack. The repeated failed login attempts, use of invalid usernames, and automated attack pattern provide strong evidence of unauthorized access attempts.

Effective log analysis and evidence handling enabled accurate identification of the threat, highlighting the importance of proper evidence analysis in SOC operations.

8. ADVERSARY EMULATION

8.1 Overview

Adversary emulation is a cybersecurity technique used to simulate real-world attacker behavior in a controlled environment. It helps organizations test their detection and response capabilities by mimicking known attack techniques based on frameworks such as MITRE ATT&CK.

Unlike penetration testing, which focuses on exploiting vulnerabilities, adversary emulation focuses on replicating attacker tactics, techniques, and procedures (TTPs). This approach enables SOC analysts to understand how attacks occur and evaluate how effectively security systems can detect and respond to them.

In this task, adversary emulation was performed to simulate a brute-force attack targeting SSH services.

8.2 Objective of Emulation

The objective of this adversary emulation was:

- To simulate a brute-force login attack
- To generate real authentication failure logs
- To test the detection capabilities of Wazuh SIEM
- To validate threat hunting and alert triage processes

8.3 Method

The attack was simulated using the following command in the Linux terminal:

ssh fakeuser@localhost

The following steps were performed:

1. Attempted to log in using an invalid username (*fakeuser*)
2. Entered incorrect passwords multiple times
3. Repeated the process to generate multiple failed login attempts

This method successfully simulated a brute-force attack scenario where an attacker tries different credentials to gain unauthorized access.

8.4 Attack Behavior Observed

During the simulation, the following behaviors were observed:

- Multiple failed SSH login attempts within a short time period
- Repeated use of invalid usernames
- Consistent source IP address (::1)
- Incremental increase in rule.firedtimes value
- Authentication failure logs recorded in /var/log/auth.log

These behaviors closely match real-world brute-force attack patterns.

8.5 Detection by Wazuh

Wazuh SIEM successfully detected the simulated attack and generated alerts based on the following:

- **Rule ID:** 5710
- **Alert Type:** SSH authentication failure
- **Log Source:** /var/log/auth.log

The system identified repeated login failures and classified them as suspicious activity. The increasing rule.firedtimes value confirmed multiple attempts, which triggered further analysis.

8.6 MITRE ATT&CK Mapping

The simulated attack was mapped to the MITRE ATT&CK framework:

- **T1110 – Brute Force**
  This technique involves systematically attempting multiple password combinations to gain access to an account.
- **T1110.001 – Password Guessing**
  A specific subtype where attackers attempt to guess passwords manually or using automated tools.

This mapping helps align the simulation with industry-standard threat intelligence.

8.7 Effectiveness of Detection

The adversary emulation demonstrated that:

- Wazuh SIEM can detect repeated authentication failures
- Alerts are generated in real time
- Attack patterns can be identified through log analysis

- SOC workflows such as triage and response can be effectively applied

This confirms that the system is capable of detecting brute-force attacks in a real-world scenario.

8.8 Conclusion

The adversary emulation successfully replicated a brute-force attack targeting SSH services. The generated logs were accurately detected and analyzed by Wazuh SIEM, validating the effectiveness of the security monitoring system.

This exercise highlights the importance of simulating attacks to test and improve detection capabilities. It also demonstrates how adversary emulation can enhance SOC readiness against real-world cyber threats.

9. POST INCIDENT ANALYSIS

9.1 Incident Overview

During the security monitoring process, a brute-force attack targeting SSH services was detected. The attack involved multiple failed login attempts using invalid usernames such as *fakeuser* and *wronguser* from a single source IP address (::1).

The activity was identified through Wazuh SIEM alerts (rule.id:5710), which indicated repeated authentication failures within a short time period. The increasing rule.firedtimes value confirmed that the attack was continuous and automated.

This incident highlights a common attack technique used by adversaries to gain unauthorized access to systems.

9.2 Root Cause Analysis (5 Whys)

A Root Cause Analysis (RCA) was performed using the 5 Whys technique to identify the underlying cause of the incident:

- Why did the attack occur?
  → Weak authentication mechanisms allowed repeated login attempts
- Why were authentication mechanisms weak?
  → No restrictions on the number of login attempts
- Why were there no login restrictions?
  → System security policies were not properly configured
- Why were policies not configured?
  → Lack of security awareness and monitoring

- Why was monitoring insufficient?
  → Absence of proactive security controls and enforcement

9.3 Root Cause

The primary root cause of the incident is identified as:

Weak authentication controls combined with lack of proper security configurations

The absence of account lockout mechanisms and monitoring allowed attackers to perform repeated login attempts without restriction.

9.4 Impact Analysis

Although the attack did not successfully compromise the system, it posed a significant risk:

- Potential unauthorized access if credentials were guessed correctly
- Increased system load due to repeated login attempts
- Exposure of system vulnerabilities
- Risk of further exploitation if not mitigated

9.5 Lessons Learned

The incident provided several important insights into system security:

- Authentication mechanisms must be strengthened
- Systems should not allow unlimited login attempts
- Continuous monitoring is essential for early detection
- Proper configuration of security tools is critical
- Regular security assessments should be conducted

9.6 Recommendations for Improvement

To prevent similar incidents in the future, the following measures are recommended:

- Implement Account Lockout Policies:
  Restrict the number of failed login attempts to prevent brute-force attacks
- Enforce Strong Password Policies:
  Use complex passwords with minimum length and character requirements
- Enable Multi-Factor Authentication (MFA):
  Add an additional layer of security beyond passwords
- Enhance Monitoring and Alerting:
  Use SIEM tools like Wazuh for real-time detection

- Regular Security Audits:
  Periodically review system configurations and logs

9.7 Continuous Improvement

Post-incident analysis is not only about identifying the cause but also about improving security posture. The insights gained from this incident should be used to update security policies, enhance detection mechanisms, and train personnel.
Continuous improvement ensures that the organization becomes more resilient against future attacks.

9.8 Conclusion

The post-incident analysis successfully identified the root cause of the SSH brute-force attack and provided actionable recommendations to strengthen system security.
By implementing the suggested improvements, the organization can significantly reduce the risk of similar attacks and enhance overall SOC effectiveness.

10. METRICS & REPORTING

10.1 Overview

Metrics and reporting are essential components of a Security Operations Center (SOC) as they help measure the effectiveness of detection and response processes. By analyzing key performance indicators (KPIs), organizations can evaluate how quickly and efficiently they identify and respond to security incidents.
Metrics provide valuable insights into SOC performance, enabling continuous improvement and better decision-making.

10.2 Key Metrics

The following metrics were used to evaluate the SOC performance during the simulated SSH brute-force attack:

- **MTTD (Mean Time to Detect):**
  1 hour
- **MTTR (Mean Time to Respond):**
  2 hours

10.3 Explanation of Metrics

- **Mean Time to Detect (MTTD):**
  This metric represents the average time taken to identify a security incident after it occurs. A lower MTTD indicates faster detection and better monitoring capabilities.
- **Mean Time to Respond (MTTR):**
  This metric measures the time taken to respond to and mitigate a detected incident. A lower MTTR reflects efficient incident handling and response processes.

10.4 Analysis

The calculated metrics indicate that the system was able to detect the brute-force attack within a reasonable timeframe. The use of Wazuh SIEM enabled continuous monitoring of logs, allowing early identification of suspicious activity.

The response time was also efficient, as the defined SOAR workflow provided a structured approach to handling the incident. Although the response actions were simulated, the workflow demonstrates how real-world automation can significantly reduce response time.

The combination of threat hunting, alert triage, and SOAR automation contributed to improved detection and response performance.

10.5 Importance of Metrics in SOC

Metrics play a vital role in improving SOC operations:

- Help identify delays in detection and response
- Provide measurable performance indicators
- Support decision-making and resource allocation
- Enable continuous improvement of security processes
- Assist in reporting to management and stakeholders

10.6 Reporting

The findings of this analysis were documented in a structured report that includes:

- Threat hunting results
- Alert triage details
- Evidence analysis
- SOAR playbook workflow
- Post-incident analysis
- Performance metrics

This report provides a comprehensive overview of the incident and demonstrates the effectiveness of SOC operations.

10.7 Recommendations for Improvement

To further improve SOC performance, the following recommendations are suggested:

- Reduce MTTD by implementing real-time alerting mechanisms
- Automate response actions to minimize MTTR
- Continuously monitor logs and system activity
- Regularly review and update detection rules
- Conduct periodic performance evaluations

10.8 Conclusion

The metrics indicate that the SOC was able to detect and respond to the simulated attack efficiently. The use of structured workflows and monitoring tools contributed to effective incident handling. Continuous monitoring and improvement of these metrics will further enhance the organization's ability to respond to future threats.

11. CAPSTONE PROJECT

11.1 Overview

The capstone project represents the integration of all SOC activities performed during this task. It demonstrates a complete end-to-end security workflow, starting from attack simulation to final reporting.

The objective of this project is to apply theoretical knowledge and practical skills to detect, analyze, and respond to a real-world cybersecurity scenario using Wazuh SIEM. It combines threat hunting, alert triage, evidence analysis, SOAR automation, adversary emulation, and post-incident analysis into a unified process.

11.2 Objective of the Capstone

The key objectives of the capstone project are:

- To simulate a real-world cyberattack scenario
- To detect malicious activity using SIEM tools
- To analyze alerts and identify threats
- To design and implement response workflows
- To evaluate SOC performance using metrics

- To demonstrate complete incident handling lifecycle

11.3 Attack Simulation

The capstone project began with the simulation of a brute-force attack targeting SSH services. This was achieved using the following command:

ssh fakeuser@localhost

Multiple incorrect password attempts were made to generate authentication failure logs. This step created a realistic attack scenario for further analysis.

11.4 Detection

The simulated attack was successfully detected using Wazuh SIEM. Alerts were generated based on:

- **Rule ID:** 5710
- **Event Type:** SSH authentication failure
- **Log Source:** /var/log/auth.log

The system identified repeated login attempts and flagged them as suspicious activity.

11.5 Alert Triage

The detected alert was analyzed and classified based on severity:

- Priority: High
- Reason: Multiple failed login attempts and brute-force pattern

This step ensured that the incident was recognized as a critical threat requiring immediate attention.

11.6 Evidence Analysis

The logs were further analyzed to gather evidence of the attack:

- Repeated login attempts from the same IP (::1)
- Invalid usernames such as *fakeuser*
- Increasing rule.firedtimes value

This analysis confirmed that the activity was malicious and not normal user behavior.

11.7 Response (SOAR Workflow)

A SOAR playbook was designed to simulate automated response actions:

1. Detect alert
2. Extract source IP
3. Validate repeated attempts
4. Classify attack

5. Block IP (simulated)
6. Create incident ticket
7. Notify administrator

This workflow demonstrates how automation can improve response efficiency in SOC operations.

11.8 Post-Incident Analysis

After handling the incident, a detailed analysis was conducted to identify the root cause and improve security measures. The analysis revealed weak authentication controls and lack of login restrictions as the primary causes.

Recommendations such as implementing account lockout policies and strengthening monitoring were proposed.

11.9 Metrics and Reporting

The performance of the SOC process was evaluated using key metrics:

- MTTD: 1 hour
- MTTR: 2 hours

These metrics indicate efficient detection and response capabilities.

11.10 Outcome

The capstone project successfully demonstrated:

- Detection of brute-force attack
- Accurate alert classification
- Effective evidence analysis
- Implementation of response workflow
- Evaluation of SOC performance

11.11 Conclusion

The capstone project validates the effectiveness of SOC operations in handling real-world cybersecurity incidents. By integrating multiple processes such as threat hunting, automation, and analysis, the system was able to detect and respond to the attack efficiently.

This project highlights the importance of a structured and proactive approach in cybersecurity and demonstrates how SIEM tools like Wazuh can enhance organizational security posture.

12. SCREENSHOTS

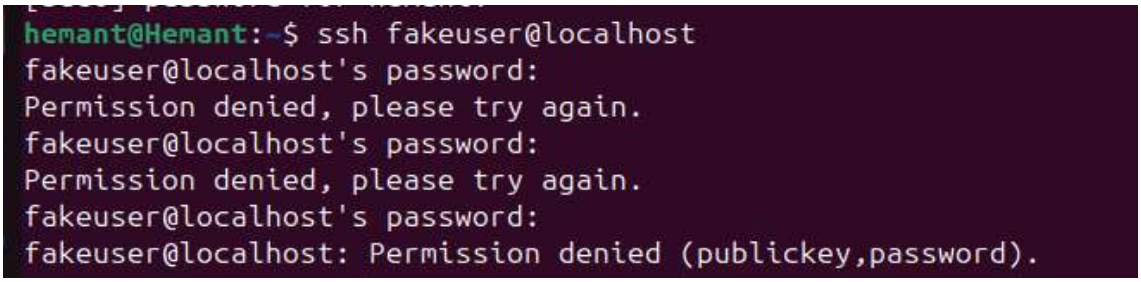

Figure 1: SSH Brute-Force Attack Simulation using Invalid User

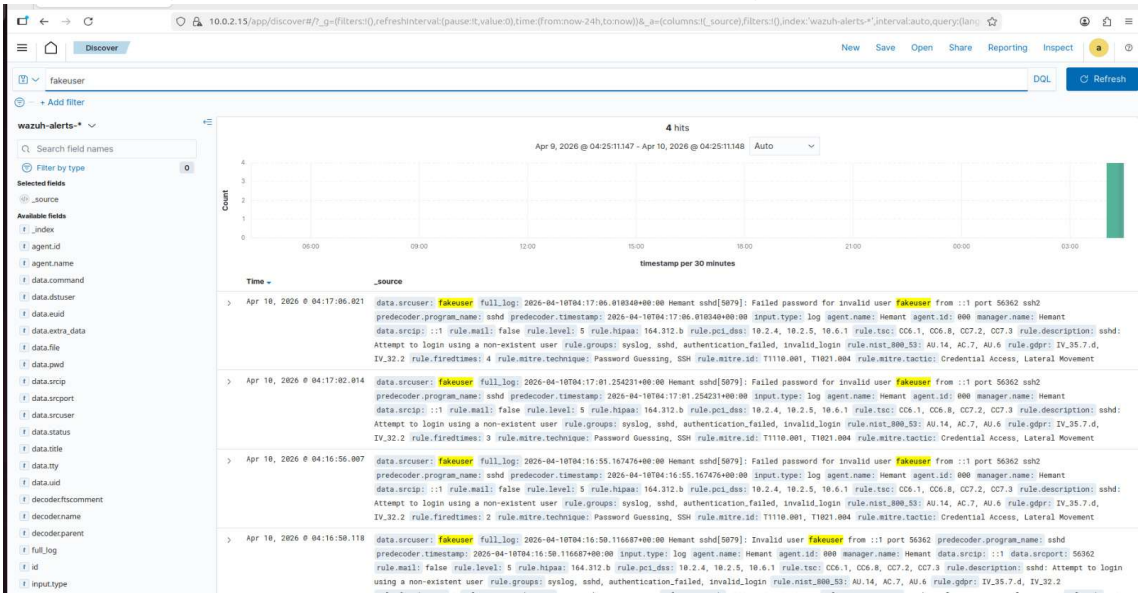

Figure 2: Wazuh Discover Logs Showing SSH Failed Login Attempts

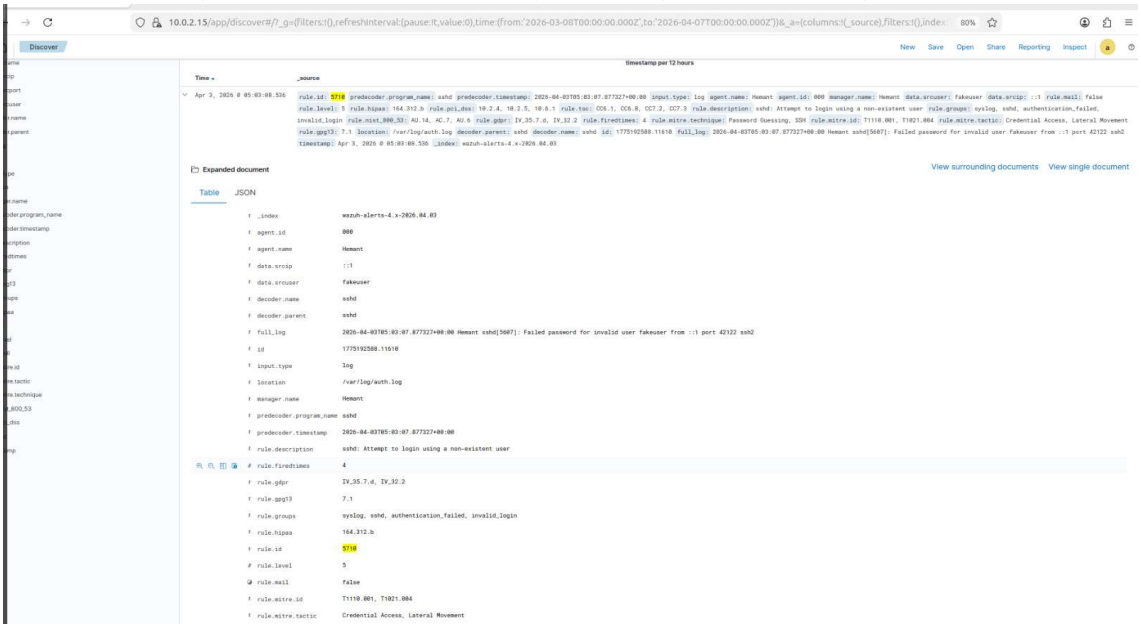

Figure 3 : Wazuh Log Details for SSH Failed Login Attempt

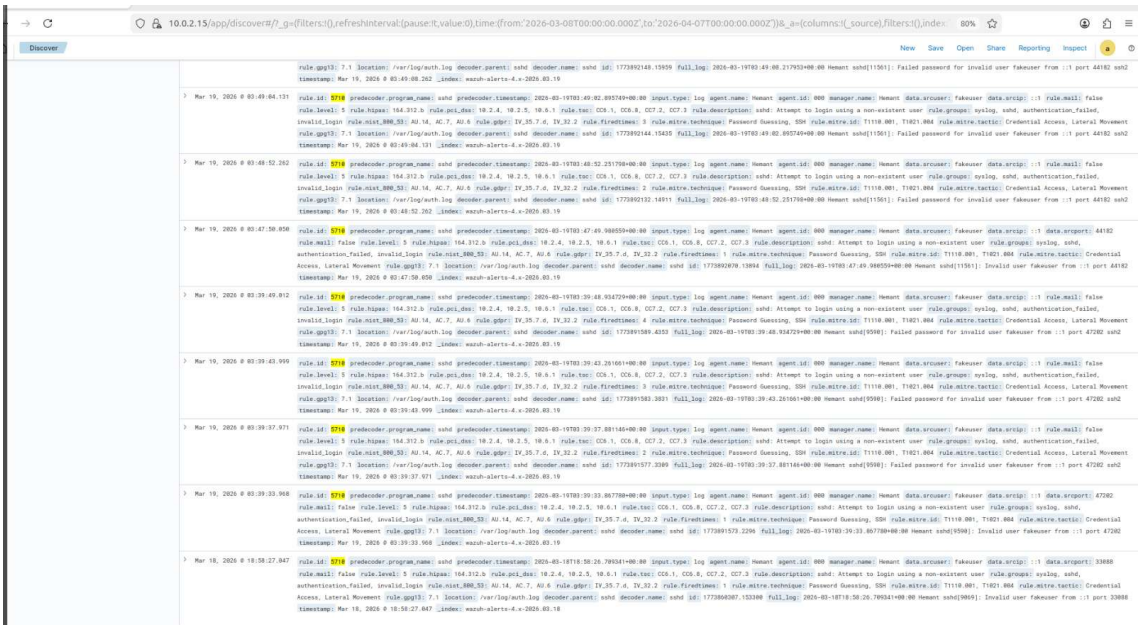

Figure 4: Repeated SSH Failed Login Attempts showing brute-force attack pattern (multiple rule.id:5710 logs for user fakeuser)

13. FINAL CONCLUSION

The Security Operations Center (SOC) workflow implemented in this project successfully demonstrated the complete lifecycle of detecting, analyzing, and responding to a cybersecurity incident. The simulated SSH brute-force attack provided a realistic scenario to apply both theoretical knowledge and practical skills.

Through **threat hunting**, suspicious activities were proactively identified by analyzing system logs. The detection of repeated failed login attempts using Wazuh SIEM confirmed the presence of a brute-force attack. The use of structured queries and log analysis enabled early identification of malicious behavior.

The implementation of a **SOAR playbook** further enhanced the incident response process by defining an automated workflow. Although simulated, the playbook demonstrated how real-world SOC environments can reduce response time and improve efficiency through automation.

Alert triage and evidence analysis played a crucial role in validating the incident. By examining log details such as source IP, invalid usernames, and increasing rule.firedtimes, the attack was accurately classified and confirmed. This ensured that the response was based on verified and reliable data.

The **adversary emulation** exercise provided valuable insights into attacker behavior. By simulating a brute-force attack, the effectiveness of the detection system was tested and validated. This highlights the importance of proactive testing in strengthening security defenses.

The **post-incident analysis** identified the root cause of the attack as weak authentication controls and lack of proper security configurations. This analysis provided actionable recommendations, including implementing account lockout policies, enforcing strong passwords, and improving monitoring systems.

Additionally, the evaluation of **SOC metrics** such as Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) demonstrated that the system was capable of detecting and responding to threats efficiently. These metrics are essential for measuring SOC performance and guiding continuous improvement.

Overall, this project highlights the importance of integrating multiple SOC processes, including threat hunting, automation, analysis, and reporting, to effectively handle cybersecurity incidents. The use of Wazuh SIEM proved to be effective in monitoring and detecting suspicious activities.